

CKD Prediction Dashboard — Detailed Feature List

- Upload multiple blood report PDFs to allow analysis across different hospital visits and time periods.
- Support both digital PDFs and scanned/image-based reports using OCR fallback for maximum compatibility.
- Automatically extract the 8 key kidney-related biomarkers required for CKD prediction.
- Capture and standardize each test's value, unit, and reference range from the report.
- Classify each biomarker as normal, borderline, or abnormal using extracted reference ranges.
- Display intuitive color-coded indicators (green, yellow, red) for quick health status understanding.
- Perform time-series trend analysis across multiple reports for each biomarker.
- Identify and display whether values are increasing, decreasing, or fluctuating over time.
- Provide an interactive dashboard with visual charts for all extracted biomarkers.
- Allow filtering to view individual biomarker trends separately for focused analysis.
- Generate CKD risk prediction using the XGBoost model based on engineered features.
- Generate CKD risk prediction using the LSTM model based on temporal sequences.
- Display risk score as a percentage along with clear color-based risk categorization.
- Provide SHAP-based explainability to show how each feature contributed to the prediction.
- Convert model explanations into simple plain-English insights for user understanding.
- Highlight the most critical abnormal biomarkers contributing to increased risk.
- Highlight rapidly worsening trends such as rising creatinine levels.
- Integrate an AI chatbot to answer user queries related to reports and medical terms.
- Implement RAG-based chatbot using extracted user report data as contextual knowledge.
- Enable context-aware answers such as explaining why a specific user's value is abnormal.
- Provide multi-visit comparison view to analyze changes between different reports.
- Ensure a clean, user-friendly interface for easy navigation and interpretation.
- Display a research disclaimer indicating the system is not a substitute for medical advice.